[pending: working title — the block-ridge arc]

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Abstract

[pending: abstract — written last, once the arc is frozen]

1 Introduction

[decision pending]

2 Data, Preparation, and Descriptive Analysis

2.1 Data

The raw data are 30-minute bars for the S&P 500 complex, gridded to a regular 30-minute frequency. The panel's extent is not set by a chosen date: **a bar exists in the panel if and only if the realized-variance source printed on it** — the same observed-print fact the availability indicators of Section 2.2 are built from, applied as the grid's own existence criterion. Under this rule the panel begins at the source's first print, in January 1998, and ends at its last, in April 2024. No weekend rule, holiday calendar, or session table is supplied anywhere: weekends, holidays, early closes, and sessions that do not yet exist are all absent from the panel for the same reason — the source did not print.

The panel's session structure therefore evolves as the market's did. The source prints around the clock from its first day — the panel's first bar is an overnight bar — but the early sessions are sparser: roughly 7,500 bars per year in 1998 (median 32 bars per trading day before 2004), growing through roughly 10,600 bars in 2003 to a stable $\approx$ 12,400 bars per year from 2005 onward, when the near-24-hour cycle (Sunday 18:30 through Friday 20:00, 48 bars per full trading day, roughly 34 of them outside regular hours) is fully live. A time-of-day slot that thickens or enters mid-sample behaves exactly like a feed going live: the rolling per-slot estimators of Section 2.2 have whatever history the slot has and accumulate it causally. Any statement about intraday structure in this panel refers to the full cycle as it stood at the time, not to the 09:30–16:00 window alone.

The existence rule replaced an earlier calendar-only filter (weekend trim plus forward fill) after that filter produced a measurable incident: bars following an early close existed on the calendar grid as forward-filled ghosts, and on one such bar — the only one in more than twenty-five thousand — two feeds' availability indicators disagreed for a single bar, with consequences documented in Section 4. On the overlapping 2005–2024 span the rule removes 4,925 such dead-session bars ($\approx 2\%$ of the calendar grid). **[pending: v3 panel counts from the harvest: grid rows, panel rows after burn-in, and scored evaluation rows after the training window and the calibration burn-in of Section 3]**

The forecasting target is realized variance. For each 30-minute bar it is computed as the sum of squared 5-minute log-returns within the bar:

$$RV_{d,j} = \sum_{i=1}^{M} r_{d,j,i}^2, \qquad M = 6, \tag{1}$$

where $r_{d,j,i}$ is the i-th 5-minute log-return inside the j-th 30-minute bar of day d. Bars are indexed sequentially as RV_t. The forecast horizon is one bar ahead: the model at bar t predicts RV_{t+1}. Throughout, "realized variance" denotes the target RV_t, and "volatility" denotes its square root; models are estimated on the square-root scale (Section 2.2). The construction in Equation (1) is the standard realized-measure definition. The heterogeneous autoregressive model of Corsi [2009] uses the trading day as its base unit; here both the target and the lag ladder are defined at 30-minute resolution.

The evaluation panel. Feature construction consumes a fixed burn-in of 3,125 bars at the start of the sample (January to June 1998), a constant of the panel construction that exceeds the longest feature lookback. This reduces the grid's 303,442 print-bars to a usable panel of 300,317 bars. The walk-forward protocol consumes a further 24,000 bars as the initial training window, and the calibration burn-in of Section 3 consumes the first evaluation window. Every arm reported in this paper is scored on the identical remaining rows; the paired bar-for-bar comparisons used throughout require this shared row index. The scored evaluation panel contains $n = 273{,}554$ bars.

Exogenous information. Beyond the target's own history, the design uses 41 exogenous channels. The channels are organised into thematic buckets that partition the columns exactly $(7 + 12 + 6 + 6 + 3 + 3 + 4 = 41)$:

- **Moments** (7): own-asset higher-order return moments: bipower variation, realized semi-variance, autocovariance, absolute, cubic, and quartic returns.
- **Liquidity** (12): trading volume, turnover (total, buy-initiated, sell-initiated), effective and quoted spreads, number of trades.
- **Market, equal-weighted** (6) and **value-weighted** (6): cross-sectional stock-factor aggregates of return moments, bipower variation, semi-variance, and absolute returns.
- **Sentiment** (3): StockTwits attention, sentiment score, and sentiment count.
- **Implied volatility** (3): VIX, VVIX, VIX3M.
- **Volatility demand** (4): options order-flow indicators (SPX open/close, all options open/close, open only).

The buckets differ in column count; liquidity contributes twelve columns and sentiment three. A per-bucket comparison therefore mixes information content with dimensionality unless it is read per feature. Availability also differs across buckets: the cross-sectional aggregates are observed on 138,357 bars, the options order-flow bucket on 54,062 bars, and the VIX family does not exist for the earliest part of the sample. Row deletion would score each bucket on a different sub-sample. Section 2.2 describes the impute-and-indicate construction used instead; it scores every bucket on the same rows.

[pending: feature accounting: a per-bucket table giving channels, the lag set applied to each, derived columns, and the reconciliation of the three circulating counts (41 channels, 359-column exogenous block, 529-column design) — the stated expansion rule of six rolling means per channel gives $41 \times 6 = 246$ and does not reconcile them; no per-column claim should be read as final until this table exists]

2.2 Preparation

This subsection specifies the preparation pipeline. The pipeline is presented in full because, measured on this panel, the representation of missing and stale data has a larger effect on QLIKE than any model-class, basis, or penalty decision reported in this paper; the magnitudes are given under "Rolling robust scaling" below. At any given bar, 36% of the exogenous design consists of forward-filled values rather than new measurements; on the sparsest channels the figure is 78%. Without availability information, an estimator cannot distinguish a filled value from a new measurement.

Overnight fills and availability indicators. The 41 exogenous channels are not all published on the near-24-hour grid. Most are US cash- or extended-session equity quantities; each prints only inside its own session. Averaged across channels, 0.638 of panel rows carry a print from the raw feed and the remaining 0.362 do not. Each channel is processed in the following order:

1. Compute a binary availability indicator from the raw feed: the indicator is 1 on bars where the feed printed and 0 otherwise.
2. Forward-fill the channel within its availability window.
3. Where the channel is structurally absent (no prior print exists), substitute a fixed in-distribution neutral value.

Every value column is accompanied by its indicator column. The indicator is computed before the fill. An earlier build of this pipeline computed the indicator after the fill, so every forward-filled bar was labelled observed. Under that build, the volatility-demand feed stopped publishing on 2023-08-31, the unlimited forward fill carried its final August print across the following eight months (8,453 bars, 3.44% of the sample), and the indicator read ≈ 1 throughout. Three further feeds (`vix`, `vix3m`, `vvix`) stopped publishing the same way on 2024-02-12. Every built cache is now gated on one check: for each channel, the mean of the constructed availability indicator must equal the fraction of rows on which the raw feed printed. The two quantities are estimates of the same number, so any gap indicates a construction error of this kind. On the uncorrected panel the check fails for 31 of 41 channels. The check requires no model, no held-out data, and no threshold tuning. We recommend it for any mixed-frequency panel in which exogenous series are filled.

The indicator also enters estimation as a regressor. Its linear coefficient absorbs any level shift between the pre- and post-availability eras, and the value column contributes only where the feed exists. Every bucket, including implied volatility and volatility demand, is therefore scored on the same 218,934-row index. The exact paired significance tests of the later sections require this shared index.

Stage 1: diurnal adjustment. Intraday volatility has a time-of-day profile. Let $s(t) \in \{1, \ldots, 48\}$ index the time-of-day slot of bar t, and let $\mathcal{W}_t = \{j : j < t,\ s(j) = s(t)\}$ be the set of the most recent $W = 20$ observations sharing that slot, with at least 5 required before the baseline is valid. For non-negative series, the baseline is the same-slot rolling mean and the adjustment divides by it:

$$\bar{RV}_{t|s} = \frac{1}{|\mathcal{W}_t|} \sum_{j \in \mathcal{W}_t} RV_j, \qquad \widetilde{RV}_t = \frac{RV_t}{\bar{RV}_{t|s}}, \tag{2}$$

Signed series (e.g. autocovariance) are divided by the same-slot rolling standard deviation instead. The window condition is $j < t$, not $j \le t$, so the baseline at bar t uses no information from bar t; the adjustment is non-anticipative. Warm-up bars pass through with baseline 1. They are consumed by the burn-in and never enter the evaluation panel.

Stage 2: the winsorized square-root target. After the diurnal adjustment, each variable class receives a fixed variance-stabilising map. The map is chosen in advance from the class's support and tail behaviour; no transform parameters are fitted. The maps are: $x \mapsto \sqrt{x}$ for squared-return and volatility-like measures, including RV itself, bipower variation, turnover, and effective spread; $x \mapsto \operatorname{sign}(x)\sqrt{|x|}$ for autocovariance; $x \mapsto x^{1/3}$ and $x \mapsto x^{1/4}$ for third and fourth moments; and $x \mapsto \log x$ for the implied-volatility family. Models are estimated on the adjusted $\sqrt{RV}$ scale. After transformation, each series is clipped to rolling 1st and 99th percentile bounds estimated on the trailing $W_{\text{clip}} = 240$ observations (about five trading days) ending at $t-1$. Winsorization is not inverted at evaluation time; only the diurnal division and the square root are inverted. Because $\mathbb{E}[\sqrt{X}]^2 \neq \mathbb{E}[X]$, squaring a forecast of $\sqrt{RV}$ gives a downward-biased forecast of RV. We therefore apply Duan's smearing correction [Duan, 1983]: add the in-sample residual variance to the squared forecast, then multiply by the stored diurnal baseline to undo the division of Stage 1:

$$\widehat{RV}_t^{\text{raw}} = \left(\hat{f}_t^{\,2} + \hat{\sigma}_\varepsilon^2\right) \cdot \bar{RV}_{t|s}^{\text{baseline}}, \qquad \hat{\sigma}_\varepsilon^2 = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2. \tag{3}$$

HAR features. The target's own history enters through trailing rolling means of the adjusted series on a base-2 geometric ladder,

$$\mathcal{L} = \{1,\ 2,\ 4,\ \ldots,\ 2^{k_{\max}}\}, \qquad \overline{RV}_t^{(k)} = \frac{1}{k} \sum_{i=0}^{k-1} \widetilde{RV}_{t-1-i}, \tag{4}$$

The rungs span roughly 30 minutes, 2.5 hours, half a trading day, 2.6 days, 13 days, and 65 trading days. The sum starts at $t-1$, so the feature at bar t contains no contemporaneous information. The ladder adapts the daily/weekly/monthly averages of Corsi [2009] to six intraday horizons. The same ladder is applied to each included exogenous channel, contributing $|\mathcal{L}| = 6$ rolling-mean columns per channel.

Rolling robust scaling. Input features to the linear estimators are standardised by subtracting a rolling median and dividing by a rolling interquartile range. Both statistics are estimated from the current training buffer and updated at every step. Two rules govern the estimation.

First, the median and IQR are estimated from observed rows only, i.e. rows on which the availability indicator is 1. If the statistics are instead estimated over forward-filled values, a series whose feed has stopped contributes identical repeated values, its rolling IQR shrinks toward zero, and the standardising division divides by a value near zero. On the uncorrected panel, the stale volatility-demand columns reached 2,260 rolling IQRs by this mechanism; the raw series did not move, and the denominator shrank.

Second, a scale is used only after 512 observed values have accumulated, and the IQR is floored at 1% of the largest scale the channel has shown. This rule exists because masked estimation fails at the start of a feed's life: with few observed rows, a low-dispersion opening window would otherwise set the scale for all subsequent bars.

The three elements — indicators computed before the fill, masked estimation, and neutral substitution — are applied together. Each element applied alone does not remove the failure. Applied together, they reduce the worst channel's maximum standardised value from 2,260.4 to 96.2 rolling IQRs. On the October 2023 window affected by the stale feed, the uncorrected panel scored QLIKE 6.37170 and the repaired panel scored QLIKE 0.12391. Every modelling effect reported in this paper lies between 0.0005 and 0.004 QLIKE. This difference in magnitude is the reason the preparation pipeline is presented before the modelling sections.

All stages are strictly causal: every quantity dated t uses only information available strictly before t.

2.3 Descriptive Analysis

[decision pending]

2.4 The OLS–HAR Benchmark

Every result in the remainder of this paper is referenced to a single fixed comparator: ordinary least squares on the six HAR features of Equation (4) plus calendar dummies, with no exogenous variables, refit at every bar. The forecasting protocol is:

1. Walk-forward with a rolling origin. The training window is the most recent $T_{\text{train}} =$ 24,000 bars (approximately 500 trading days at 48 bars per day). The window slides forward by one bar per prediction.

2. The model is refit at every bar. The refit is exact and computationally feasible because the sliding-window normal equations admit rank-1 up-dating (add the entering bar) and down-dating (remove the exiting bar).

3. The benchmark is read from the ridge path at $\alpha = 0$; it is therefore exact OLS.

4. Each forecast is mapped back to raw variance through Equation (3).

5. Forecasts are scored by QLIKE [Patton, 2011],

$$L_{\text{QLIKE}} = \frac{1}{N} \sum_{t=1}^{N} \left[r_t - \log r_t - 1 \right], \qquad r_t = \frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}}, \tag{5}$$

computed on the full 218,934-bar evaluation panel. QLIKE is used for three reasons: it is scale-invariant in the level of volatility; it is strictly consistent for the conditional variance; and it penalises under-prediction more heavily than over-prediction, which matches the direction in which forecast errors are more costly for this target. QLIKE values are not comparable across preprocessing regimes: on the diurnally adjusted panel of the reference study, the same class of models scores several tenths of a QLIKE unit lower than on a raw, unadjusted panel. Every comparison in this paper is therefore made within one preprocessing regime, and every table is labelled with the panel it was computed on.

[pending: the benchmark table: per-bar OLS-on-HAR QLIKE and secondary metrics (out-of-sample R^2, Mincer–Zarnowitz α/β, MSE, MAE) from the unification campaign's `a0` arm on the frozen base-2 panel — the single benchmark every comparison in this paper is measured against]

Differences in QLIKE at the fourth decimal place require a significance assessment. Every comparison against this benchmark is therefore accompanied by a Diebold–Mariano test [Diebold and Mariano, 1995] on the per-bar loss differential. The test procedure is:

1. Compute the per-bar QLIKE loss for each of the two arms on the shared row index, and take the per-bar difference.

2. Estimate the long-run variance of the loss differential with a Newey–West estimator.

3. Apply the small-sample correction of Harvey et al. [1997] to the test statistic.

Where several arms are compared simultaneously, we report the Model Confidence Set of Hansen et al. [2011]. Every arm is scored on the identical row index, so all of these tests are paired bar-for-bar comparisons.

Hyperparameter discipline. No quantity anywhere in this paper is selected using evaluation loss. Every hyperparameter that is tuned is tuned *causally*: the penalty (or penalty vector) is re-selected periodically from a grid fixed in advance, by minimizing validation error on a forward split of the current training window — fit block, embargo, validation tail — so that no forecast's configuration uses information from that forecast's future. Every hyperparameter that is not tuned is a fixed constant, set before the campaign was scored and reported with its arm: the fixed-penalty arms exist precisely as controls for what tuning contributes, and the structural constants of the design (training-window length, re-selection cadence, winsorization quantiles, block widths, estimation-window lengths) are conventions of the panel and estimator construction, disclosed where they are defined and never optimized against any loss.

3 A Calibrated Second-Moment Correction

3.1 Why the Back-Transform Needs a Second Moment

Every model in this paper is fit on the preprocessed scale. Realized variance RV_t is diurnally adjusted and then passed through the square-root transform, so the regression target is $y_t = \sqrt{\widetilde{RV}_t}$, where $\widetilde{RV}_t$ denotes the diurnally adjusted variance at bar t. Evaluation is performed on the raw variance scale. This is because QLIKE — the primary loss — is a strictly consistent scoring rule for the conditional variance, not for any transform of it. A forecast issued in square-root space must therefore be mapped back to raw space before it is scored.

The naïve inverse squares the forecast and reverses the diurnal adjustment. Squaring does not commute with conditional expectation: by Jensen's inequality, $\mathbb{E}[\sqrt{X}]^2 \leq \mathbb{E}[X]$, with equality only in the degenerate case. Decompose the target as $y_t = \hat{y}_t + \varepsilon_t$, where $\hat{y}_t$ is the model's forecast of y_t and ε_t is a conditionally mean-zero innovation. Then

$$\mathbb{E}\left[y_t^2 \,\middle|\, \mathcal{F}\right] = \hat{y}_t^2 + \mathbb{E}\left[\varepsilon_t^2 \,\middle|\, \mathcal{F}\right], \tag{6}$$

where $\mathcal{F}$ is the conditioning information set. The conditional mean of the squared target exceeds the squared forecast by the conditional second moment of the square-root-scale innovation. A forecast that is well calibrated in $\sqrt{RV}$-space is, once squared, biased low in RV-space, and the size of the bias equals the residual variance of the model being evaluated.

Two facts about this bias determine the design. First, QLIKE penalizes under-prediction more heavily than over-prediction, and the bias of the uncorrected squared forecast is downward. Second, the bias equals each model's own residual variance, so it differs across models. The back-transform therefore includes an explicit second-moment term, and the estimator of that term is specified as part of the evaluation procedure.

3.2 The Calibrated Estimator

The correction is a two-component causal calibration, re-estimated for every evaluation window on the window before it. Fix the chunked walk-forward grid: the backtest is split into contiguous evaluation windows indexed $k = 0, 1, \ldots$, tiling the panel in row order. The inputs for window k are three per-bar arrays over its bars: the forecasts $\{\hat{y}_t\}$ on the transformed (square-root, diurnally adjusted) scale, the diurnal baseline factors $\{B_t\}$, where B_t is the factor that reverses the Stage-1 adjustment at bar t, and the raw realizations $\{RV_t^{\text{raw}}\}$. The evaluation-scale target is the *unwinsorized* transformed realization, reconstructed exactly from the persisted raw target,

$$y_t^{\text{raw}} = \sqrt{RV_t^{\text{raw}}/B_t}. \tag{7}$$

Mean calibration. The forecast is passed through one affine map per window,

$$m_t = \hat{a}_k + \hat{b}_k\, \hat{y}_t, \qquad (\hat{a}_k, \hat{b}_k) = \arg\min_{a,\, b} \sum_{s\, \in\, k-1} \big(y_s^{\mathrm{raw}} - a - b\, \hat{y}_s\big)^2, \tag{8}$$

the Mincer–Zarnowitz regression of the evaluation-scale target on the forecast, estimated by ordinary least squares over the valid bars of window $k-1$ and applied to every bar of window k. The map absorbs the level and slope miscalibration the fit-scale construction leaves behind, including the attenuation induced by training on the winsorized target. For a window whose predecessor is missing from the harvest — a mid-panel gap, which a complete harvest does not contain — the map falls back to the identity $(\hat{a}_k, \hat{b}_k) = (0, 1)$; every such window is flagged in the harvest accounting.

Conditional second moment. Let $e_s = y_s^{\mathrm{raw}} - m_s$ denote window $k-1$'s residuals against its own calibrated mean, as applied there. The second-moment term for window k is the scalar

$$\hat{\sigma}_k^2 = \frac{1}{N_{k-1}} \sum_{s\, \in\, k-1} e_s^2, \tag{9}$$

one constant per window, estimated on the previous window, where N_{k-1} counts the valid bars entering the mean. Positivity is trivial — a mean of squares — and the estimator does not extrapolate: no value of window k's forecasts can move it. For a window whose predecessor is missing from the harvest, it falls back to the in-window scalar mean of $(y_t^{\mathrm{raw}} - m_t)^2$; every such window is flagged in the harvest accounting.

Calibration burn-in. The first evaluation window of each arm is the calibration burn-in, parallel to the feature construction's burn-in at the start of the panel: it estimates the two maps for its successor and is not scored — none of its bars enters any loss, any join, or any calibration diagnostic. Scoring begins at the second window. Every scored forecast is therefore computable at its own bar, with no exceptions.

The conditional-variance record. Two bar-conditional parameterizations of the second moment were implemented and rejected on measurement. An affine form, $\hat{\sigma}_t^2 = \max(\hat{c}_0 + \hat{c}_1 \hat{y}_t^2, 0)$, fit on the previous window's squared residuals, admits zero variance on quiet bars and lets the raw forecast collapse: under it, twenty bars carried 99% of one arm's pooled QLIKE. A log-linear form, $\hat{\sigma}_t^2 = \exp(\hat{d}_0 + \hat{d}_1 \hat{y}_t^2)\, \hat{S}$, with $\hat{S}$ the Duan [1983] smearing factor for the log-retransformation, is structurally positive but extrapolates exponentially at bars whose $\hat{y}_t^2$ lies outside the previous window's fitted range: the second moment reached 10^{44}, inflating every arm's loss and collapsing the Diebold–Mariano differentials toward zero. A bar-conditional second moment therefore requires a model with controlled extrapolation, which is the reserved learned-correction slot for this section; the contract uses the scalar of Equation (9).

A shrunk conditional second moment. The two rejected forms fail for the same reason: neither is anchored to the estimator that works. A third form removes that failure mode by construction. Let

$$z_t \;=\; \log\!\big((y_t^{\mathrm{raw}} - m_t)^2\big),$$

and predict z_t by ridge regression on a standardized design, fit on the rolling window and refit at every bar. The penalty is selected causally on the same embargoed forward split used for every other hyperparameter in this paper, from a grid that includes $\alpha = \infty$; at that grid point the fitted value is the training-window mean of z and the assembled variance is *exactly* the scalar $\hat{\sigma}_k^2$ of Equation (9). The scalar contract is therefore nested as the infinite-shrinkage limit, and conditioning is adopted only where the forward split prefers it. At finite α the variance is

$$\hat{v}_t \;=\; \exp(\hat{z}_t)\, c_k, \qquad c_k \;=\; \frac{\text{mean of } (y^{\text{raw}} - m)^2 \text{ on window } k-1}{\text{mean of } \exp(\hat{z}) \text{ on window } k-1},$$

so each window's variance *level* is pinned to the scalar estimator's level and conditioning can only redistribute variance across bars within a window. No floor, clip, or additive constant appears anywhere in the construction.

Measured against the benchmark on the shared panel, the conditional contract is the largest single effect in this study: applied to the benchmark's own residuals it improves QLIKE by -0.00453 at DM -12.0, and applied to the paper's best mean model by -0.00394 at -9.4. Both are an order of magnitude larger than any increment produced by changing the conditional mean.

Which state carries the second moment. That result invites the reading that the exogenous panel predicts forecast uncertainty. A control arm rejects it. Conditioning the variance on the backbone alone — the autoregressive ladder and calendar dummies, with no exogenous column — improves the benchmark by -0.00535 at DM -15.3, and beats the exogenous-state version head to head (-0.00082 at -2.9). The predictable part of the squared forecast error is therefore carried by the volatility level itself, which is what the realized-variance estimator's own measurement error scales with, and not by the exogenous state. The conditional second moment is a large and robust effect about heteroskedasticity in the target; it is not evidence that the exogenous buckets carry distributional information beyond the level, and this paper does not claim that they do.

Back-transform. Both members of the scored pair are mapped to the raw scale as

$$\widehat{RV}_t^{\text{raw}} = \left(m_t^2 + \hat{\sigma}_k^2\right) B_t, \qquad RV_t^{\text{raw}} = \left(y_t^{\text{raw}}\right)^2 B_t, \tag{10}$$

where the right-hand identity is Equation (7) inverted: the realization is the persisted raw target itself, never reconstructed from the winsorized fit target. The traditional scalar estimator of Duan [1983] is the same scalar form with the identity mean map and in-window fit-scale residuals: the contract differs from Duan smearing in exactly two ways — the calibrated mean, and previous-window residuals against that mean on the evaluation scale.

The full scoring procedure for one evaluation window is:

1. *Inputs.* The per-bar arrays $\{\hat{y}_t\}$, $\{B_t\}$, and $\{RV_t^{\text{raw}}\}$ over windows $k-1$ and k; reconstruct y^{raw} by Equation (7).

2. *Calibration.* Estimate $(\hat{a}_k, \hat{b}_k)$ by Equation (8) and $\hat{\sigma}_k^2$ by Equation (9) on window $k-1$'s valid bars, with the stated fallbacks.

3. *Back-transform.* Map every bar of window k to the raw scale with Equation (10), producing the per-bar pairs $\left(\widehat{RV}_t^{\text{raw}}, RV_t^{\text{raw}}\right)$.

4. *Loss.* Evaluate QLIKE on the raw-scale pairs,

$$\text{QLIKE} = \frac{1}{N}\sum_{t=1}^{N}\left(\frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}} - \log\frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}} - 1\right), \tag{11}$$

where the mean is taken over the bars at which both members of the pair are positive; bars failing that condition are excluded from the mean.

Because $\hat{\sigma}_k^2$ is a mean of squares and $m_t^2 \geq 0$, the calibrated forecast $\widehat{RV}_t^{\text{raw}}$ is positive whenever B_t is and window $k-1$'s residuals are not identically zero; the positivity condition of the exclusion rule then binds only through the realization.

Three implementation choices define the estimator.

One calibration per window, estimated on the previous window. The three scalars $(\hat{a}_k, \hat{b}_k, \hat{\sigma}_k^2)$ are window-level constants, re-estimated at every window boundary from the window before it. The correction is piecewise-constant across the panel on the transformed scale; on the raw scale it varies bar by bar through the baseline factor B_t. Relative to the squared calibrated forecast, the correction is largest at the bars where m_t^2 is smallest — the quiet bars.

Causal, hence deployable. Every estimated quantity applied in window $k \geq 1$ is a function of data through window $k-1$ only, and windows are processed in strictly ascending order. The map is a quantity a deployed forecaster could have computed at t: the scored object is a real-time forecast, not a retrospectively corrected one. Scoring begins after the calibration burn-in, so there are no exceptions.

Residuals of the model itself, against the unwinsorized target. The residuals entering Equation (9) are the evaluated model's own out-of-sample forecast errors from the preceding window, measured against the unwinsorized evaluation-scale target of Equation (7) — not training-set residuals, not the residuals of any reference model, and not fit-scale residuals. The correction is therefore model-specific: an arm with larger residual variance receives a larger upward correction, so the correction is part of each arm's forecast, not a shared post-processing constant. With this choice the decomposition of Equation (6) holds for the evaluation target itself, tails included. One bias remains and is stated plainly: the *coefficients* of each arm are estimated on the winsorized fit target, so the forecast $\hat{y}_t$ itself carries the fit-scale clipping. That is a modeling choice rather than a scoring defect; the mean calibration of Equation (8) absorbs its first-order attenuation, it is shared by every arm, and it cancels to first order in comparisons.

3.3 What Downstream Inherits

Every forecast in this paper passes through Equation (10) before any QLIKE number is computed. Every downstream comparison **[pending: enumerate the downstream analyses by their final section labels (bucket attribution, block structure, model classes) once those sections stabilize]** scores its arms on raw-scale pairs produced by exactly this map, with exactly these window and residual choices.

Because the calibration pair and the second-moment coefficients are model-specific, the correction does not shift all arms by a common constant. It inflates the forecasts of an arm with larger residual variance by more, and it straightens the forecasts of a miscalibrated arm by more. Under QLIKE's asymmetric penalty, this interacts with each arm's calibration in a way that a shared correction would not. A comparison run under a different convention — an uncalibrated squared forecast, a scalar in-window smear, or no correction at all — would be a comparison of different forecast objects, and it would not necessarily induce the same ranking.

That comparison is run, restricted to exactly three conventions on the identical persisted per-bar arrays and the identical exclusion rule. Convention *none* is the naïve inverse of Section 3.1: the squared forecast with the diurnal adjustment reversed, $\widehat{RV}_t^{\text{raw}} = \hat{y}_t^2 B_t$, with no second-moment term. Convention *Duan* is the in-window scalar smear — the Duan [1983] special case identified after Equation (10) — with one scalar per evaluation window, the mean of that window's own squared fit-scale residuals, added to the squared forecast before the baseline reversal. Convention *contract* is the calibrated stack of Equations (8)–(10), the convention every headline number in this paper inherits. Table 1 reports each completed arm's pooled QLIKE under the three conventions, with within-convention ranks. The rank-stability summary is the Kendall rank correlation between the contract ranking and each alternative's: $\tau = 0.637$ against *none* and $\tau = 0.918$ against *Duan*.

Table 1: The benchmark under the three back-transform conventions. The full per-arm table is Appendix Table 6; Kendall τ of each alternative ranking against the contract ranking over all arms: none 0.637, traditional Duan 0.918.

	No correction	Traditional Duan	Contract
OLS on HAR + calendar (benchmark)	0.24097	0.23165	0.23353

The correction is treated as part of the evaluation contract: it is frozen here, before the first comparison, and no downstream section revisits it. Any future change to the second-moment model — including the learned correction contemplated for this slot — would constitute a change of contract, and would require re-scoring every comparison that inherited the old one.

4 Where the Information Lives, and What Extracts It

The reference point for this section is the per-bar least-squares fit of the HAR ladder plus calendar dummies, QLIKE 0.23353. This section measures two quantities: the change in QLIKE when each exogenous feature bucket is added to that backbone, and the overlap between the buckets' individual contributions. The buckets are defined in **[pending: cross-ref to the panel/descriptive section defining the buckets]**. The estimator is held fixed at the least-squares class throughout this section. The same features under a richer hypothesis class are treated in **[pending: cross-ref to the linear-vs-nonlinear section]**.

Estimator. Every arm in Table 2 is *minimum-norm least squares*: among all coefficient vectors that minimize the training-window sum of squares, the unique one of minimal ℓ_2 norm, computed through the pseudoinverse of the window gram at the standard machine-precision cutoff. This is the $\alpha \to 0^+$ limit of ridge regression, it is defined at any design rank, and it carries no penalty and no tuned quantity of any kind — the strongest attribution claim an unpenalized estimator supports. Each arm is refit at every bar over a rolling window of 24,000 bars (500 trading days). On a full-rank design (the benchmark's 52 columns) minimum-norm least squares coincides exactly with ordinary least squares. An earlier pass of this campaign used the plain normal-equations solve, whose solution is arbitrary in near-null directions of the design; two single-window incidents documented below trace to that path, and its results are archived. Because the pseudoinverse is defined at any rank, the estimator requires no rank guards; the only design intervention retained is the go-live participation rule of Section 2.2, which is a statement about data existence, not numerics.

Protocol. The attribution is computed as follows.

1. *Arms.* The incumbent arm's design matrix contains the HAR ladder columns and the calendar dummies. Each bucket arm's design matrix contains those backbone columns plus the bucket's own columns, expanded through the same moving-average ladder and availability indicators as the joint arm. The joint arm's design matrix contains the backbone columns plus all 41 exogenous series ($\approx$ 526 columns after the moving-average ladder and availability indicators).

2. *Fit.* Each arm is fit by the rolling estimator above and produces one out-of-sample forecast per bar.

3. *Loss.* Each arm is scored by per-bar QLIKE on the shared panel (n = 218,934 thirty-minute bars, identical rows in every cell). The reported QLIKE is the mean of the arm's per-bar losses.

4. *Increments.* Δ is the arm's QLIKE minus the incumbent's QLIKE. Δ_{own} is the arm's QLIKE minus the same estimator's own no-exogenous baseline (0.13445).

5. *DM test.* For each arm, the per-bar loss differential against the incumbent is $d_t = \ell_t^{\text{arm}} - \ell_t^{\text{inc}}$. The Diebold–Mariano statistic [Diebold and Mariano, 1995] is the t-ratio of $\bar{d}$ to its Newey–West HAC standard error, with the Harvey–Leybourne–Newbold small-sample correction [Harvey et al., 1997]. Negative t favours the arm.

6. *Overlap.* The overlap ratio is

$$R = \frac{\sum_b \Delta_{\text{own}}(b)}{\Delta_{\text{own}}(\text{joint})},$$

   the sum of the single-bucket within-class increments divided by the joint arm's within-class increment, both measured against the same estimator's no-exogenous baseline.

4.1 Per-Bucket Gains

Table 2 reports the causally tuned ridge arm on each of the exogenous buckets, on the shared panel, against the shared incumbent.

Table 2: Bucket attribution: literal per-bar OLS on each exogenous bucket, from the unification campaign. **Panel:** frozen base-2 panel, diurnally adjusted, $n = 218{,}934$ thirty-minute bars, identical rows in every cell; incumbent QLIKE 0.24050 (per-bar OLS on HAR + calendar). **Estimator:** exact OLS ($\alpha = 0$), rolling window 24,000 bars (500 trading days), refit every bar by exact rank-one updates. Columns participate in a window only when their feed went live before the window starts and printed within it; masked columns are disclosed per chunk. Δ is against the incumbent; the within-class increment coincides with Δ because the no-exogenous design is exactly the incumbent under literal OLS.

Bucket	QLIKE	Δ vs. incumbent	DM t
all_features (joint)	0.23444	+0.00091	+1.6
moments	0.23089	-0.00264	-5.7
liquidity	0.23471	+0.00118	+3.3
implied_vol	0.23363	+0.00010	+0.4
vol_demand	0.23375	+0.00022	+1.2
sentiment	0.23253	-0.00100	-5.2
market_ew	0.23357	+0.00004	+0.1
market_vw	0.23303	-0.00050	-1.2
(no exogenous = incumbent)	0.23353	—	—

On the full 1998–2024 panel, two single-bucket arms reject equal predictive accuracy with the incumbent in the bucket's favor: moments (-0.00264, t = -5.7) and sentiment (-0.00100, -5.2). One leans the same way without significance (market VW, -1.2); three are indistinguishable from the incumbent (market EW, +0.1; implied volatility, +0.4; volatility demand, +1.2). The liquidity arm is significantly worse (+0.00118, +3.3), and the joint arm — all exogenous columns in one minimum-norm fit — is statistically indistinguishable from the incumbent (+0.00091, +1.6): even the strongest unpenalized estimator extracts essentially nothing from the stacked design, while the same columns under the per-block shrinkage of Sections 4.5–6 drive the paper's best forecasts. That contrast — information present, unpenalized joint extraction null, shrunk extraction strong — is this table's central message.

A disclosure attaches to the market-aggregate rows. In earlier passes of this campaign both aggregates appeared catastrophically harmful. The cause was a single evaluation window (December 2016 – February 2017) in which a pair of availability-indicator columns, differing only through pre-2015 history, decayed to exact collinearity as that history left the rolling window; the linear solver does not signal a nearly singular system, and the contaminated coefficients produced forecast errors that one window's correction then spread across all of its bars. The per-bar solve verification of the final design (Section 2) closes this failure mode; under it the aggregates score as reported in Table 2.

4.2 The Same Buckets under a Competent Estimator

Table 2 answers what an unpenalized estimator recovers. It is not the same question as what the columns contain. Each bucket design is therefore re-run under two penalized estimators, both choosing their penalty by the causal rule of Section 4.5: ridge, and an elastic net whose mixing parameter is *free* — the tuner selects ℓ_1 share as well as penalty strength at every retune, so the amount of selection is chosen by the estimator rather than fixed by the author. Table 3 reports both.

Table 3: Each bucket design under two causally tuned penalized estimators. **Panel:** the frozen panel of Section 2, identical rows in every cell; benchmark QLIKE 0.23353. DM t is against the benchmark; negative favours the arm. The final column is the paired Diebold–Mariano statistic of the elastic net against ridge *on the same design*, positive meaning ridge is the better of the two.

	Ridge		Elastic net		Enet vs. ridge
Design	QLIKE	DM t	QLIKE	DM t	DM t
moments	0.23089	-3.6	0.23134	-2.5	+1.9
liquidity	0.23431	+0.8	0.23453	+0.9	+0.9
market (EW)	0.23354	+0.0	0.23382	+0.4	+1.0
market (VW)	0.23296	-0.9	0.23361	+0.1	+2.4
sentiment	0.23357	+0.1	0.23389	+0.5	+2.0
implied vol.	0.23362	+0.2	0.23401	+0.6	+1.5
vol. demand	0.23397	+0.7	0.23422	+0.9	+1.6
all exogenous (joint)	0.23040	-4.4	0.23116	-2.0	+1.2

Two readings follow, and they are the reason this section and the estimator comparison belong together.

First, *the joint design is now the best design, not the worst.* Under minimum-norm least squares the joint arm was indistinguishable from the benchmark (Table 2); under a causally tuned ridge it is the strongest arm in the table (0.23040, DM -4.4), stronger than any single bucket including the best of them. The ordering of the two tables is not a refinement of the same result but a reversal of it: with an estimator that shrinks, more exogenous columns produce more signal; with one that does not, they produce none. What Table 2 measures is the cost of forgoing shrinkage, and that cost is larger than everything the buckets individually contain.

Second, ridge attains a lower pooled loss than the elastic net in every information set. All eight paired comparisons favour ridge, and three do so at conventional significance (market VW +2.4, sentiment +2.0, moments +1.9). That ordering, however, is not yet a statement about the two *families*, for the reason given next.

Where the elastic net's deficit comes from. Because the elastic net selects its penalty causally at every retune, each scored bar can be attributed to the penalty in force when it was forecast, and the deficit can be located rather than merely totalled. It does not spread across the sample. On the 72.1% of bars whose governing penalty was interior to the grid ($\alpha \leq 10^{-3}$), the elastic net *beats* tuned ridge by 3.9×10^{-4} (DM -6.70, favouring the elastic net in all eight designs). The entire pooled deficit is contributed by the 27.9% of bars whose governing penalty sat at the *top point of the grid*, $\alpha = 10^{-2}$, where it loses by 2.5×10^{-3} (DM $+6.85$, in all eight designs and in 18 of 24 years). The difference between the two subsets is 2.9×10^{-3} at $z = 7.8$.

A result carried by a grid endpoint invites the explanation that the grid is too small. Under the correspondence between the elastic net's parameterization and a ridge penalty — an ℓ_2 weight of $N\alpha(1-\rho)$ on the window gram, with $N = 24{,}000$ and ρ the mixing share — the elastic net's largest expressible ℓ_2 weight is $24{,}000 \times 10^{-2} \times 0.75 = 180$, while the tuned ridge arm draws from a grid reaching 1,000 and selects that top value in 41.3% of its own retunes. The elastic net cannot express the shrinkage ridge chooses, and it loses on the bars where it has run out of grid.

That explanation is testable and it is false. Re-running every design with the penalty grid extended to $\alpha = 1$ — which reaches 6 to 18 times ridge's selected weight at every mixing value, and contains the original grid as a subset — makes the elastic net *substantially worse*, not better. On the moments design the extended-grid arm loses 2.9×10^{-3} to its own narrow-grid twin (DM $+7.01$) and 3.3×10^{-3} to tuned ridge (DM $+6.68$); on the joint design, 1.7×10^{-3} (DM $+4.56$) and 2.4×10^{-3} (DM $+3.46$). Giving the estimator more room to shrink causes it to shrink worse.

This locates the asymmetry precisely, and it is not about how much shrinkage each family can express. Both tuners pin the tops of their grids — ridge in 41.3% of retunes, the elastic net in a comparable share — because on a 125-bar validation tail neither can resolve the penalty. What differs is the consequence. For ridge, the loss is monotone and flat in $\log \alpha$ near its optimum (Section 4.5.1), so selecting too large a penalty costs almost nothing: over-shrinking a dense coefficient vector degrades gracefully. For the elastic net, a larger penalty is partly an ℓ_1 penalty, and over-penalizing in that direction removes columns outright. The same selection error is benign in one family and destructive in the other. Selection loses here not because the data prefer dense coefficients, but because the price of a hyperparameter mistake is asymmetric — and a short causal validation window guarantees such mistakes.

What the trajectories do and do not show. The selected mixing share is bimodal — 36.7% of retunes at pure lasso, 32.5% at the minimum 0.25, mean 0.64 — and the same shape appears in every bucket. It is tempting to read this as the estimator declining to select. Two measurements argue against that reading. Holding α fixed, the mixing share accounts for only 4.2×10^{-4} of the deficit, significant in one of eight designs and reversing sign at $\alpha = 10^{-4}$; and the selected mixing share is unrelated to the volatility regime (Spearman -0.013 against the preceding window's realized volatility, $p = 0.23$). The mixing trajectory is a symptom of a validation tail too short to identify the penalty, not the channel through which the loss is incurred.

Nor is ridge exempt from that instability: it pins an endpoint of its own grid in 58.1% of retunes, against 43.3% for the elastic net. The asymmetry this section can support is therefore narrower than a family verdict — both estimators are choosing badly on a 125-bar tail, and ridge's bad choices happen to be inexpensive while the elastic net's are not.

4.3 Overlap

The overlap ratio — the sum of single-bucket increments divided by the joint increment — cannot be computed from Table 2, because the joint arm's increment is positive: at this dimensionality the unshrunk joint fit's variance cost exceeds any information gain, so the joint OLS arm does not measure the buckets' combined information. The available overlap evidence comes from shrunk estimators: an earlier battery under a causally tuned ridge measured $R = 2.08$ over its seven buckets, and that battery's causally tuned LightGBM arm $R = 2.24$ — individual gains summing to roughly twice the joint gain, under both estimator classes, on the prior convention. **[pending: overlap ratio recomputed from shrunk joint fits under the paper's frozen convention — the block estimators of Sections 4.5–6 provide the joint fits]**

4.4 The Joint Bucket

The joint `all_features` arm — every exogenous column in one minimum-norm fit — has QLIKE 0.23444, statistically indistinguishable from the incumbent (+1.6). The same columns under the per-block shrinkage of Sections 4.5 and 5 produce the paper's best forecasts (0.22111 at -7.7), so what the joint row measures is the cost of forgoing shrinkage, not the absence of information.

The estimator choice matters here, and its history is part of the record. Under the earlier plain normal-equations solve, this arm appeared catastrophically worse than the incumbent; the deficit traced to solution components in near-null directions of the design — large mutually canceling coefficients on nearly duplicate availability-ladder columns, which a one-bar feed desynchronization on a forward-filled post-early-close ghost bar (December 24, 2020, the only such disagreement in over twenty-five thousand bars) turned into a forecast of 77.8 against a target of 0.26, whose residual the window-level correction then spread across its whole window. The minimum-norm solution suppresses near-null components by construction, and the dead-session grid rule removed the ghost-bar class; under the two repairs the arm scores as reported.

An independent every-bar robustness design was run on a different panel: rolling 1,000-trading-day window; HAR block exactly unpenalized via Frisch–Waugh–Lovell, with only the exogenous block penalized; scored on a fixed COVID-inclusive window from January 2018 to May 2025, $n = 74{,}934$ bars. On that panel, every bucket rejects equal predictive accuracy against HAR-only (the weakest, `vol_demand`, at $t = -6.2$), and the joint set rejects equal predictive accuracy against every single bucket, at $t = -13.0$. The bucket ranking differs across the two panels: `implied_vol` has the largest single-bucket increment on the COVID-inclusive window, `moments` in Table 2.

An earlier draft asserted that the 90% model confidence set [Hansen et al., 2011] over the feature sets is the singleton containing the joint bucket. The per-bar loss series required to compute a model confidence set were not retained for the battery, so no set was computed, and the claim is not repeated here. **[pending: battery MCS over the feature sets once per-bar losses are persisted]** Table 2 supports three statements: three buckets individually beat the incumbent at the 5% level and two more lean the same way without significance; the unshrunk joint fit is worse than the incumbent and than every single bucket;

and extracting the buckets' combined information therefore requires the shrinkage machinery the remainder of the paper develops.

4.5 A Dense but Weak Signal: Lasso versus Ridge

The preceding subsections established that the exogenous panel carries information beyond the HAR lag structure of Corsi [2009], that no single bucket carries much of it, and that the estimator decides whether any of it is recovered. This subsection holds the information set fixed and varies only the penalty of the linear estimator. Two estimators are compared: the lasso [Tibshirani, 1996], which applies an ℓ_1 penalty and sets a subset of coefficients to exactly zero, and ridge [Hoerl and Kennard, 1970], which applies an ℓ_2 penalty and shrinks all coefficients toward zero without setting any to zero. The result of the comparison is a property of how the predictive coefficients are distributed across the columns of the design. In this section that property is measured directly: ridge attains lower out-of-sample QLIKE than the lasso on the full design (Section 4.5.1); the lasso zeroes 304 of 523 exogenous columns on the full basis, of which 235 are dead or constant availability indicators (Section 4.5.2); and five principal components retain 72% of the feature variance while losing essentially all of the predictive signal (Section 4.5.2). The section calls this coefficient structure *dense but weak*: many nonzero coefficients, none individually large. Section 4.5.3 defines the estimator built for this structure: a two-block ridge with penalty $\alpha = 1$ on the HAR-of-target block and $\alpha = 100$ on the HAR-of-all-features block, with no selection step.

4.5.1 The Head-to-Head

Design. The two penalized linear estimators are run on the full feature design — every exogenous column at once — on one shared out-of-sample panel of $n = 218{,}934$ thirty-minute bars, with identical rows in both cells; nothing varies between the two arms except the penalty.

Estimators. The design matrix X has one row per bar in the current training window and one column per feature: the HAR ladder of the target, the calendar features, and the full exogenous block of 523 columns. Let y be the vector of targets over the same window, and let x_s denote row s of X. The two estimators solve, at each refit,

$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^{\top}\beta\right)^2 + \alpha\,\|\beta\|_2^2,$$
$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^{\top}\beta\right)^2 + \lambda\,\|\beta\|_1,$$

where the sums run over the bars of the training window and the penalties α and λ are held constant between the periodic re-selections described below. The ℓ_1 penalty produces a coefficient vector supported on a subset of columns; the ℓ_2 penalty produces a coefficient vector with all entries nonzero.

Walk-forward protocol. The panel is computed as 100 walk-forward chunks over a rolling 500-trading-day (24,000-bar) training window. Every arm is refit *every bar*: at each bar the oldest bar leaves the window, the newest enters, and the coefficients are re-estimated before the next forecast is issued. Each arm re-selects its own penalty periodically and causally: every 250 solves, the current training window is split forward into a fit block, a 25-bar embargo (covering the longest moving average entering the features), and a 125-bar validation tail, and the penalty value minimizing validation MSE on that tail is adopted. No forecast's hyperparameters ever see its own future. The lasso arm is solved as an exact every-bar online homotopy: the solution and active set at each bar are obtained by exactly updating the previous bar's solution for the one-row change in the window, not by a warm-started batch solver terminated at a tolerance, so the head-to-head is not confounded with optimization error. The ridge arm runs on an exact rank-one update path.

Comparison protocol. Both arms are compared against the shared incumbent of Section 2: per-bar OLS on HAR plus calendar features with no exogenous inputs, at the benchmark QLIKE of 0.23353. For each arm, let ℓ_t^{arm} and ℓ_t^{inc} be the per-bar QLIKE losses [Patton, 2011] of the arm and the incumbent at bar t, and let $d_t = \ell_t^{\mathrm{arm}} - \ell_t^{\mathrm{inc}}$. The Diebold–Mariano statistic [Diebold and Mariano, 1995] is the sample mean $\bar{d}$ divided by its Newey–West HAC standard error, with the small-sample correction of Harvey et al. [1997] applied; a negative t favors the arm. **[decision pending]**

Result. The comparison is run on the full design — the HAR ladder, the calendar features, and the full exogenous block — with both estimators under the same causal tuning protocol, so that neither family carries an informed penalty choice the other lacks (Table 4). Ridge (0.23040, DM -4.4 against the incumbent) beats the lasso (0.23134, -2.0), and the elastic net — the interpolating family at the battery's mixed-penalty grid — lands between the two (0.23056, -2.6): the loss is monotone in how densely the penalty spreads the coefficients. Keeping every column and shrinking produces a lower loss than selecting a subset, when every family chooses its penalty by the same causal rule.

Table 4: Ridge versus lasso on the full design, out-of-sample QLIKE with the Diebold–Mariano t-statistic against the OLS–HAR incumbent in parentheses. Unification campaign, $n = 218{,}934$ thirty-minute bars, identical rows in every cell; negative t favors the arm. Both arms re-select their penalty every 250 solves from the battery grids on a forward validation split.

	Ridge	Elastic net	Lasso
Causally tuned	0.23040 (-4.4)	0.23056 (-2.6)	0.23134 (-2.0)
Incumbent (per-bar OLS–HAR)		0.23353	

A penalty-level sweep locates where the tuned arms operate. Holding the ridge family fixed and varying only the constant, QLIKE falls monotonically in α across the tested grid: 0.23219 at $\alpha = 0.1$, 0.23159 at 0.3, 0.23083 at 1, 0.23003 at 3, and 0.22909 at 10, still decreasing at the grid's edge; the tuned-ridge arm, which re-selects from a grid reaching $\alpha = 1000$, lands below every fixed point. The penalty *level* moves the loss by more than the penalty *family* does at any fixed level, and the wide design rewards heavier shrinkage than the $\alpha = 1$ default — the direction the block penalties of Section 4.5.3 take further. One caveat: all DM statistics are computed against the incumbent, a third model, so the ridge-minus-lasso differences themselves carry no significance statement yet. **[pending: pairwise ridge-vs-lasso DM (or the incumbent-cancelling bound) on the full design, so the head-to-head margin is tested rather than pointed at]**

4.5.2 The Diagnosis: Dense, Not Sparse

The lasso's own solution path shows what ℓ_1 selection does on this design. On the full basis the lasso zeroes 304 of 523 exogenous columns. Of those 304, 235 — 77% — are dead or constant availability indicators. Most of the lasso's selection therefore removes columns that carry no signal. Removing such columns does not lower QLIKE relative to ridge, because ℓ_2 shrinkage makes the same columns small without making them zero. Where the ℓ_1 penalty zeroes live columns, QLIKE rises: the margin by which ridge beats the lasso is carried by the live columns the lasso zeroes, not by the dead ones.

A principal-component check measures whether the spread of coefficients across columns is a low-dimensional structure in a rotated basis. Truncating the design to five principal components retains 72% of the feature variance and loses essentially all of the predictive signal. The signal therefore does not lie in the high-variance directions of the feature covariance. The measured structure is: (i) no small subset of columns concentrates the signal — the only columns the lasso can zero without raising QLIKE are dead or constant ones; and (ii) no small set of principal directions concentrates it. The coefficient structure has many small nonzero entries in the coordinate basis and is not low-rank in the data's covariance geometry. This is the structure the section labels dense but weak.

4.5.3 The Two-Block Ridge

The head-to-head leaves two facts that a single ridge penalty cannot satisfy at once. First, penalizing the HAR-plus-calendar design alone raises QLIKE relative to plain OLS (the battery's no-exogenous control arm: ridge $t = +5.0$ against the incumbent): the HAR columns are few, well-conditioned, and individually strong, so any shrinkage on them is pure bias. Second, the wide exogenous design requires heavy shrinkage — that is the head-to-head's result. A scalar penalty applies the same α to both blocks: a value large enough for the hundreds of weak exogenous columns over-shrinks the HAR columns, and a value small enough for the HAR columns under-shrinks the exogenous columns. The resolution is a penalty that differs by block.

The section's estimator is a **two-block ridge**: one jointly solved least-squares problem with a block-diagonal Tikhonov penalty,

$$(\hat{\beta}_1, \hat{\beta}_2) = \arg\min_{\beta_1,\beta_2} \|y - X_1\beta_1 - X_2\beta_2\|_2^2 + \alpha_1\|\beta_1\|_2^2 + \alpha_2\|\beta_2\|_2^2, \qquad (\alpha_1, \alpha_2) = (1, 100).$$

Block 1 (X_1) is the HAR ladder of the target, at $\alpha_1 = 1$. Block 2 (X_2) is the same HAR averaging construction applied to *every* exogenous feature — HAR-of-`all_features`, the full wide basis — at $\alpha_2 = 100$. The two blocks are not fit separately and stacked; the coefficients are estimated in one solve. The per-block penalties reduce to a single ridge by column scaling: with $\tilde{X} = \left[X_1/\sqrt{\alpha_1} \;\; X_2/\sqrt{\alpha_2} \right]$ and $\gamma = (\sqrt{\alpha_1}\,\beta_1,\ \sqrt{\alpha_2}\,\beta_2)$,

$$\|y - X_1\beta_1 - X_2\beta_2\|_2^2 + \alpha_1\|\beta_1\|_2^2 + \alpha_2\|\beta_2\|_2^2 = \left\|y - \tilde{X}\gamma\right\|_2^2 + \|\gamma\|_2^2,$$

so the two-block problem is an ordinary ridge with unit penalty on the rescaled design, and $\hat{\beta}_j = \hat{\gamma}_j/\sqrt{\alpha_j}$ for $j = 1, 2$. **[pending: how the $(\alpha_1, \alpha_2) = (1, 100)$ pair was fixed — grid/validation provenance, and its sensitivity]**

The estimator matches the measured coefficient structure clause by clause. The signal is spread across many columns, so block 2 keeps all of its columns: nothing is selected. No individual exogenous coefficient is large, so block 2 is shrunk two orders of magnitude harder than block 1, and the fit sums many attenuated contributions. The HAR columns are few and strong, so block 1 is penalized only at $\alpha_1 = 1$; setting $\beta_2 = 0$ reduces the model to ridge at $\alpha = 1$ on the backbone, which differs from the OLS–HAR incumbent of Section 2 only by that nominal penalty. The estimator contains no selection step and no tuned dial: the penalties are fixed constants, not optimized quantities.

The two-block ridge scores QLIKE 0.22941, with Diebold–Mariano -9.0 against the OLS–HAR incumbent. Under the documented convention (penalties $1/3 \times 10^3$, 250-day window) the same structure scores 0.22775 at -9.6. These are the first scored runs of this exact specification. **[pending: DM of the two-block ridge against single-penalty ridge on `all_features` — computed from the same per-bar losses at harvest]**

The next two sections extend this estimator one block at a time. Section 5 identifies the nonlinearity the trees use, writes it as an explicit product block, and estimates a three-block ridge. Section 6 adds a block of factor lead–lag transmission features and estimates the final four-block model. In every extension, each block keeps all of its columns, each block carries its own fixed penalty, and nothing is selected.

5 Nonlinearity and the Product Block

Section 4.5 established the linear results: the exogenous signal is dense and weak, uniform heavy shrinkage is the measured linear response, and the two-block ridge is that response. This section measures what nonlinear models add. It has three parts. First, gradient-boosted trees beat the best linear arm on every feature set, by a small margin (Section 5.1). Second, decompositions of the tree fits attribute the bulk of that margin to additive-plus-pairwise structure, and the pairwise part to products of columns the panel already contains, gated on the session clock. Third, those products are written down as explicit columns, appended to the linear design, and shrunk as a block of their own (Section 5.2). The section ends with the resulting three-block ridge, the second rung of the block ladder this paper builds (two blocks in Section 4.5, four in Section 6).

5.1 The Tree Edge

Protocol. The evidence is the causal-tuning battery of Section 4: five estimators fully crossed with the nine feature buckets on the shared panel ($n = 218{,}934$ thirty-minute bars, identical rows in every cell). The two nonlinear arms are the gradient-boosted tree ensembles LightGBM [Ke et al., 2017] and XGBoost [Chen and Guestrin, 2016]. The linear arms are the penalized regressions of Section 4.5. Every arm, tree and linear alike, reselects its own hyperparameters on the same periodic causal schedule: each reselection uses only data available before the reselection date, so no forecast's configuration uses information from that forecast's future. The shared comparator is the per-bar OLS–HAR incumbent (Section 2.4).

Results. LightGBM has the lowest QLIKE on all nine feature sets. Ridge, the best linear arm, is second on all nine. The ranking reverses on no bucket. On the richest set, `all_features`, LightGBM scores 0.12604 and ridge scores 0.12788, a model-class gap of 0.00184; across the nine sets the tree-minus-linear gap ranges from 0.00184 to 0.00358. An exact sign test over the nine sets gives $p = 0.0039$. A bound-based pairwise test resolves all nine matched comparisons in the tree's favour after Holm adjustment; the test uses the fact that the incumbent cancels from the tree-minus-linear loss differential, so the pairwise standard error is bounded by the sum of the two arms' standard errors with no assumption on their dependence. The `all_features` pair is the weakest at $|t| \geq 2.19$, $p \leq 0.029$; the other eight range from $|t| \geq 3.73$ to $|t| \geq 5.91$. The model-class gap is roughly a quarter of the information effect of Section 4.

The function-class premium at fixed information. The battery's no-exogenous `baseline` row uses the incumbent's information set exactly: HAR lags and calendar features, nothing else. On that row the causally tuned LightGBM scores 0.13118 ($t = -5.6$) and XGBoost scores 0.13163 ($t = -5.5$). Measured against the base-2 OLS ladder at 0.13332, the premium from relaxing the linear functional form is 0.00215. Measured against the causally tuned ridge on the same features (0.13445), the premium is 0.00327. We carry the smaller figure, 0.00215, forward. On this same row the three linear arms are the only arms in the battery with higher QLIKE than the incumbent.

Dispersion of the improvement. The Diebold–Mariano statistics [Diebold and Mariano, 1995, Harvey et al., 1997] against the incumbent on `all_features` order the two arms opposite to their QLIKE levels: ridge reaches $t = -19.9$ and LightGBM reaches $t = -15.4$. The DM statistic is the mean per-bar loss improvement divided by the standard error of that improvement, so a larger $|t|$ at a smaller mean improvement implies a smaller bar-to-bar dispersion of the improvement: ridge's per-bar improvement has the smaller dispersion, the tree's the larger mean.

[decision pending]

Where the nonlinearity lives. Three decompositions of the tree fits were computed in an earlier phase of this project.

Interaction order. A functional-ANOVA decomposition of a fitted depth-eight boosted tree attributes approximately 55% of its explainable variance to additive terms, 9% to pairwise interactions, and 36% to a higher-order remainder concentrated in no identifiable interaction. Within the additive part, the HAR block accounts for 96%.

Saturation at pairwise order. On the deployed residual-stack panel of the earlier build (n = 194,934; levels not comparable to the battery), an expressivity ladder holds the base model fixed and varies only the correction stage. A purely additive correction scores 0.12757. Adding bilinear pairwise interactions scores 0.12108. Nonlinear pairwise shapes score 0.12080. The deployed bagged additive-plus-pairwise stage (a GA2M in the sense of Lou et al. 2013) scores 0.12033. An Optuna-searched unrestricted-depth XGBoost correction stage scores 0.12094, which is worse than fitting no correction at all (0.12081). The ladder improves monotonically up to pairwise order; the unrestricted stage does not improve on the pairwise stages.

Distillation into products. On an earlier build of the same panel family, a penalized linear base scores 0.12516 and a residualized boosted-tree correction improves it to 0.12414. Adding to the base a single hand-written interaction — the HAR lags multiplied by a late-day session dummy, six columns — scores 0.12457, which recovers 58% of the tree correction's improvement. The full set of HAR-by-six-regime interactions scores 0.12453, or 62%. The session dummy in the recovered term marks the late-day bars at which the change of sign in HAR's persistence coefficient is documented in Section 2.

The tree experiments therefore yield a specification statement: **the panel's recoverable nonlinearity is a sparse set of pairwise products of columns the design already contains**. The rest of this section constructs that set of products as an explicit feature block.

5.2 The Product Block

The construction enumerates candidate products, selects a subset, and appends the subset to the linear design as ordinary columns, read by the same rolling ridge machinery as every other block. The interaction study that developed the block settled four design choices: which products, reselected how often, shrunk how hard, and refit how often. Its numbers were computed on the interaction-study panel — 218,934 out-of-sample bars, 2007–2024, a 250-day rolling window, HAR reference QLIKE 0.13275 — which shares its rows with the battery panel but not its window or backbone convention. The levels below are therefore not comparable to the battery tables and are quoted only for the design decisions they settle. **[pending: confirmation that the paper's production product block is exactly this construction (candidate space, selection window, column scaling) once the block is re-run under the paper's panel convention]**

Construction. The block is built in four steps. The constants below are those frozen in the campaign executor, `src/unification.py`, the code that runs the campaign's product arms; where the interaction study's description and the executor differ, the executor's construction is the one stated here, because it is what ran.

1. *Candidate set.* The base set is the target's HAR ladder, the 41 exogenous value channels at trailing windows of 1, 32, and 512 bars, and the four session-clock columns (the open, close, and overnight indicators and the hour of day) — 133 columns on the interaction-study panel. The candidate set is every pairwise product of two base columns, including squares: 8,911 candidate products.

2. *Selection target.* A backbone-only ridge — the HAR-plus-calendar block at penalty $\alpha = 1$, refit every bar — is walked forward over the first $2W$ rows of the panel, where W is the arm's rolling fit window: 24,000 bars (500 days) under the stated convention of Section 5.3, 12,000 bars (250 days) under the documented convention. The selection target is that ridge's out-of-sample residual on rows $[W, 2W)$.

3. *Scoring and selection.* Each base column is centered over rows $[W, 2W)$. Each candidate is scored by the absolute correlation, over those rows, between the product of its two centered parents and the selection target. The $k = 100$ candidates with the largest scores are kept. Selection runs once; the set is never reselected; and every arm carrying the block is evaluated only from row $2W$ onward, so no scored bar overlaps the selection window.

4. *Column scaling.* Each kept product is formed bar by bar as the product of its two parent columns as they stand in the panel (uncentered, carrying the causal prescaling of Section 2). The product column is then divided by its own trailing rolling standard deviation over a W-bar window, lagged one bar and requiring at least 1,000 observations, with the scale floored from below at 0.1 times the per-bar cross-column median of the product columns' rolling standard deviations — the floor bounds the inflation a low-dispersion trailing window can produce, the failure mode the scaling discipline of Section 2 guards against. Rows before the first valid scale, all inside the first window and before any scored bar, reuse the first valid value. A product with zero dispersion over its entire trailing window is set to zero on those bars. Every product value at bar t is a function of data available at bar t.

Set size. The interaction study's set-size ladder is unimodal: accuracy peaks at $k = 25$–100 of 8,911 and degrades by $k = 400$. The linear channel of Section 4.5 improved monotonically in the number of columns; the interaction channel peaks and degrades. The block uses $k = 100$.

Reselection cadence. Two arms were compared with identical machinery and identical coefficient refits, differing only in when selection runs. The static arm selects the 100 products once, on the first selection window (February 2001 through October 2003 in calendar time), and keeps them for the rest of the sample. The dynamic arm reselects the product set every month. The dynamic arm loses: at $k = 100$ it trails the static arm at DM $t = -3.21$ under the study's original scaling, and the sign is negative at every healthy specification tested. The dynamic set replaces 24–31% of its members per month. The product set is selected once, eighteen years before the end of the sample, and is never reselected.

Block penalty. Products of centered features have variance on the order of the product of their parents' variances, so the product block and the base block have columns on incommensurate scales. Under a single shared penalty the product block's measured contribution is unstable and can be negative; an early +43% headline for the block was an artifact of a design clip that was standing in for the missing block-wise regularization. Under a separate penalty, with the product block shrunk an order of magnitude harder than the linear block, the hundred frozen products add $\Delta R^2 = +0.0067$ over the full 516-column linear base — a +35% relative improvement in residual-space accuracy, DM $t = +2.71$. Lighter product penalties eliminate the gain.

Coefficient refit cadence. The comparison holds the frozen block and the solver fixed and varies only how often the coefficients are re-estimated. At a monthly coefficient refit, the block's QLIKE increment is indistinguishable from zero (DM $t = +0.52$). At a daily refit, the identical frozen block's residual-space increment is +0.0108, and the raw-scale reconstruction gives $\Delta\text{QLIKE} = -0.00093$, DM $t = +3.56$. The selected products are predominantly fast pairs built from one-bar quantities. The identity of the products is frozen; their coefficients are refit frequently.

Assembled, the product block is one hundred columns with four measured properties — sparse, frozen, heavily shrunk, frequently refit — and it converts the specification statement of Section 5.1 into a feature block the paper's estimator can carry.

5.3 The Three-Block Ridge

The model this section ends on is the two-block ridge of Section 4.5 with the product block appended as a third block. Each block carries its own ridge penalty inside one rolling solver:

$$\hat{y}_t \;=\; \underbrace{X_t^{\text{HAR}}\beta_1}_{\alpha_1=1} \;+\; \underbrace{X_t^{\text{exog}}\beta_2}_{\alpha_2=100} \;+\; \underbrace{X_t^{\text{prod}}\beta_3}_{\alpha_3=1000}, \tag{12}$$

where block one is the target's own HAR ladder at $\alpha_1 = 1$; block two is the linear exogenous block of Section 4.5 at $\alpha_2 = 100$; and block three is the frozen product block at $\alpha_3 = 1000$. The estimator is implemented by column scaling. Each column in block b is multiplied by $\sqrt{\alpha_1/\alpha_b}$, and a single ridge with penalty α_1 is solved on the scaled design; a coefficient $\tilde{\beta}$ on a scaled column corresponds to $\beta = \sqrt{\alpha_1/\alpha_b}\,\tilde{\beta}$ on the raw column, and the shared penalty $\alpha_1\|\tilde{\beta}\|^2$ on the scaled coefficients equals the per-block penalties $\alpha_b\|\beta\|^2$ on the raw coefficients. After scaling, the three-block estimator runs in the same exact rank-one rolling least-squares path as every other linear model in this paper, refit at every bar. Setting $\beta_3 = 0$ recovers the two-block model exactly.

Each penalty restates a measured result. $\alpha_1 = 1$: Section 4.5 measured that shrinking the low-dimensional, fully informative HAR block raises loss, so the backbone is left essentially unpenalized. $\alpha_2 = 100$: the exogenous signal is dense and weak, and heavy uniform shrinkage was the measured optimum. $\alpha_3 = 1000$: the interaction study measured that the product block's contribution is positive and stable only when it is shrunk an order of magnitude harder than the linear block; the one-order-of-magnitude gap between α_2 and α_3 is the ratio the study's own ladder selected on its panel. No penalty was chosen by grid search on the evaluation loss.

The campaign executor runs the three-block arm under two conventions. The stated convention is Equation (12) at a 24,000-bar (500-day) rolling fit window: penalties $(\alpha_1, \alpha_2, \alpha_3) = (1, 100, 1000)$. The documented convention pins $(1, 3 \times 10^3, 3 \times 10^4)$ at a 12,000-bar (250-day) window. Under the stated convention the three-block ridge scores QLIKE 0.22111, with Diebold–Mariano -7.7 against the OLS–HAR incumbent; under the documented convention it scores 0.22652 at -10.6. The product-block *increment* — the paired per-bar comparison of the three-block against the two-block ridge on the same rows — is ΔQLIKE = -0.00006 at DM -0.2 under the stated convention, and -0.00123 at -7.1 under the documented one; under causal per-block tuning, -0.00055 at -1.4. **[decision pending]**

The product block does not carry all of the tree's fit: the functional-ANOVA remainder and the expressivity ladder's residual gap sit above pairwise order, and the block contains pairwise products only. What the block does carry runs in a linear estimator — one solver, three penalties, refit every bar. The next section adds the fourth block.

6 Transmission: Factors and Cross-Factor Structure

The three blocks assembled so far are contemporaneous: the target's own history, the exogenous panel, and the frozen pairwise products are all functions of data observable at the current bar. This section adds a fourth block built on the panel's cross-sectional structure. The construction has three steps. First, a factor frame is estimated from the eigenstructure of the panel's first training window. Second, a one-bar cross-correlation field is estimated on that frame and its antisymmetric part is retained: a matrix stating, for every ordered pair of factors, the degree to which the first leads the second at a one-bar delay. Third, that field is applied as a linear operator — no clustering, ranking, or partitioning step — and the resulting columns, together with the factor scores themselves, enter the ridge as a fourth block.

The construction is directional by design, and this section reports that its measured value is *not.* Section 6.4 substitutes the symmetric part of the same cross-correlation for the antisymmetric part, substitutes the undecomposed cross-correlation for both, and removes the operator entirely; none of these substitutions changes the block's contribution by a detectable amount, while the operator columns alone are the weakest member of the family. We therefore describe the block's value as cross-factor co-movement carried by well-scaled factor levels, and we retain the directional construction only because it is the one the columns are actually built from. The lead–lag reading, which motivated the construction and which an earlier draft asserted, is withdrawn on the evidence in Section 6.4.

6.1 Twenty Natural Factors

The factor construction uses no rotation, no supervision, and no tuning. Write $x(t) \in \mathbb{R}^{133}$ for the panel's base-feature vector at bar t. Of the 133 columns, $p = 106$ are live, non-degenerate directions — sample standard deviation above the degeneracy threshold over the frame window — indexed by the set J with $|J| = p = 106$. Fix the frame window

$$\mathcal{T}_0 \;=\; \{\, W+1, \ldots, 2W \,\}, \qquad W = 24000 \text{ bars (500 trading days)},$$

the window immediately following the first W-bar fit window (February 2001 through October 2003 in calendar time — the dot-com bust, the September 2001 disruption, and the pre-recovery trough; the frame is estimated in one regime and applied across all that follow, a disclosure the split-half replication results below address directly). Standardize each live column by its frame-window moments,

$$z_j(t) \;=\; \frac{x_j(t)-\mu_j}{\sigma_j}, \qquad \mu_j = \frac{1}{W}\sum_{t\in\mathcal{T}_0} x_j(t), \qquad \sigma_j^2 = \frac{1}{W}\sum_{t\in\mathcal{T}_0}\big(x_j(t)-\mu_j\big)^2,$$

and form the correlation matrix of the live columns over the frame window,

$$R_{jk} \;=\; \frac{1}{W}\sum_{t\in\mathcal{T}_0} z_j(t)\, z_k(t), \qquad j,k\in J,$$

with eigendecomposition

$$R\, v_k \;=\; \lambda_k v_k, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p.$$

The factor frame is the set of top-q eigenvectors $v_1,\ldots,v_q$ with $q = 20$, and the factor score $G_i(t)$ is the projection of the bar-t feature vector onto eigenvector i, re-standardized by its own frame-window moments:

$$G_i(t) \;=\; \frac{1}{s_i}\Big(\sum_{j\in J} v_{i,j}\, z_j(t) \;-\; m_i\Big), \qquad i=1,\ldots,q,$$

where m_i and s_i are the mean and standard deviation of the raw projection $\sum_{j\in J} v_{i,j}\, z_j(t)$ over $\mathcal{T}_0$. Every quantity in the construction — μ_j, σ_j, v_i, m_i, s_i — is estimated once, from feature data alone, on $\mathcal{T}_0$, and the frame is then **frozen permanently** before any forecasting begins. Because the frame is estimated from X and never from the target, objects built on it can be validated by split-half replication — estimating the same object independently on the two halves of the sample and correlating the two estimates — without contamination from the signal. The lead–lag field built on this frame has the highest split-half replication of any object in the study (Section 6.2).

The count of $q = 20$ was carried as a convention and was not optimized. Three statistics of the same first-window spectrum bracket it. The participation ratio of the eigenvalue spectrum,

$$\mathrm{PR} \;=\; \frac{\big(\sum_{k=1}^{p}\lambda_k\big)^2}{\sum_{k=1}^{p}\lambda_k^2} \;\approx\; 18,$$

an effective dimension count, puts the span at effectively $\approx$ 20 dimensions. The raw Marchenko–Pastur edge,

$$\lambda_+ \;=\; \big(1+\sqrt{p/T}\big)^2,$$

the largest eigenvalue expected from a correlation matrix of independent noise at dimension p and sample size T, evaluated at the nominal sample size $T = |\mathcal{T}_0|$, leaves 22 of the 106 live eigenvalues above the noise level. And the linear frame is stable across halves of the 18-year sample, while every nonlinear geometric object built from the factors (curvature, spectra of interaction forms, cluster partitions) fails split-half replication. Twenty is the round cover of the 18–22 bracket. Section 6.2 reports the experiment that later varied the count directly.

One refinement of the Marchenko–Pastur count matters for everything that follows. Bar data is serially dependent. For factor i with lag-k autocorrelation $\rho_i(k)$, the integrated autocorrelation time

$$\tau_i \;=\; 1 + 2\sum_{k\geq 1}\rho_i(k)$$

is the factor by which serial correlation reduces the effective number of independent observations; its median across the factor scores is 175 bars (≈ 3.6 days). Substituting $T_{\text{eff}} = T/\tau$ for T deflates the effective sample to $T_{\text{eff}} \approx 68$ per dimension and raises the dependence-adjusted noise edge λ_+ to 5.04. Only **5** eigenvalues exceed that edge, so only 5 eigenvectors are individually identified. This does not mean the mid-spectrum is noise: objects built on the mid-spectrum replicate split-half at +0.6 to +0.99, and noise replicates at zero. The correct statement is that the data pins the subspace the mid-spectrum eigenvectors span but not the basis inside it: the individual eigenvectors are not identified, while their span is stable under split-half replication. Section 6.2 tests the implied prediction directly in the width experiment: performance depends on covering the span, not on the choice of directions inside it.

6.2 One-Bar Lead–Lag Features

The lead–lag field. Two controls precede the estimate, since the deterministic intraday timetable could manufacture artificial lead–lag. First, each factor score is deseasonalized per time-of-day slot: with $s(t)$ the slot of bar t and $\mu_{i,s}$, $\sigma_{i,s}$ the mean and standard deviation of G_i within slot s, the deseasonalized score is

$$\tilde{G}_i(t) \;=\; \frac{G_i(t) - \mu_{i,s(t)}}{\sigma_{i,s(t)}},$$

so that no score carries a diurnal mean or variance profile. Second, an explicit clock control is computed: the field implied by the deterministic time-of-day component alone is estimated separately, and its share of the dynamic flow is -0.003. Let C be the lag-one-bar cross-correlation matrix of the deseasonalized factor scores,

$$C_{ij} \;=\; \operatorname{corr}\big(\tilde{G}_i(t-1),\, \tilde{G}_j(t)\big),$$

and let

$$A \;=\; \frac{C - C^\top}{2}$$

be its antisymmetric part: $A_{ij} > 0$ states that factor i leads factor j at one bar more than j leads i. The pairwise lead–lag statement is the whole content of the object; no clustering, ranking, or partitioning is performed anywhere in this construction — A (estimated causally as D below) enters the feature map directly as a linear operator. Methods that summarize such a field into a partition or a global ordering (the lead–lag clustering literature; **[pending: cite]**) were applied only as diagnostics, and both fail on it: a signed-clustering partition replicates below its permutation null and a global ranking replicates at $\tau \approx -0.06$, because a field that is 96.5% curl has neither a partition nor a ranking to recover. Lead–lag at one bar is therefore the correct description at the level of factor *pairs*, while the global structure is a circulation — transmission around cycles, not a hierarchy from leaders to laggards.

The field is the most replicable object in the study. Split-half replication is the correlation of the vectorized upper triangles of the two half-sample estimates $A^{(1)}$ and $A^{(2)}$,

$$\rho_{1/2} \;=\; \operatorname{corr}\Big(\big(A^{(1)}_{ij}\big)_{i<j},\; \big(A^{(2)}_{ij}\big)_{i<j}\Big);$$

the dynamic flow replicates at $\rho_{1/2} = +0.992$. A third control addresses construction arithmetic: features built from trailing moving averages induce a mechanical lag-one cross-correlation, a closed-form expression for that induced component explains 6% of the field's variance, and the residual after removing it still replicates at +0.992.

Three properties describe the field. First, its Hodge decomposition splits the antisymmetric flow into a gradient part — the component consistent with a single ranking of the factors by a potential $s \in \mathbb{R}^q$ — and a cyclic (curl) part:

$$A \;=\; A^{\text{grad}} + A^{\text{curl}}, \qquad A^{\text{grad}}_{ij} \;=\; s_i - s_j, \qquad s \;=\; \arg\min_s \sum_{i<j} \big(A_{ij} - (s_i - s_j)\big)^2,$$

whose least-squares solution on the complete graph is the row mean $s_i = \frac{1}{q}\sum_j A_{ij}$ (up to an additive constant), with $A^{\text{curl}} = A - A^{\text{grad}}$ the residual. The curl share

$$\frac{\|A^{\text{curl}}\|_F^2}{\|A\|_F^2} \;=\; 96.5\% :$$

the field is **96.5% curl**. Second, the field estimated on regular trading hours and the field estimated overnight agree at +0.967 (correlation of vectorized upper triangles, as above). Third, its hub structure is: the turnover and tail-activity factors lead the implied-volatility factors, and the implied-volatility factors lead the realized-volatility factors, at the one-bar (half-hour) lag.

From field to feature. The feature construction applies the field as a linear operator, estimated causally on a trailing 504-day window and re-estimated quarterly. Fix the trailing length $L = 504$ trading days and the refresh cadence $\Delta = 63$ trading days. At each refresh time τ on the grid, the lag-one-bar cross-correlation of the factor scores is estimated over the trailing window and antisymmetrized,

$$C^{(\tau)}_{ij} \;=\; \operatorname{corr}_{\tau-L<t\le\tau}\big(G_i(t-1),\, G_j(t)\big), \qquad D^{(\tau)} \;=\; \frac{C^{(\tau)} - C^{(\tau)\top}}{2},$$

and the operator in force at bar t is piecewise constant between refreshes,

$$D(t) \;=\; D^{(\tau(t))}, \qquad \tau(t) \;=\; \max\{\,\tau \le t : \tau \text{ on the refresh grid}\,\},$$

with $D(t) = 0$ before the first full trailing window exists. The transmission columns are

$$\hat{G}_j(t) \;=\; \sum_{i=1}^{q} D_{ij}(t)\, G_i(t - 1 \text{ bar}),$$

one column per factor: column j is the next-bar motion of factor j as predicted by the flow through it; each column is then scaled by its own trailing standard deviation, computed causally. The antisymmetric part is used, and not the full lag-one cross-correlation, on the reasoning that the symmetric part is persistence content the moving-average ladder already spans. That reasoning is what the construction encodes; it is tested in Section 6.4, where the symmetric part and the undecomposed cross-correlation are both substituted for D directly, and it does not survive the test.

The design choices above — lag one, quarterly re-estimation — were fixed in an earlier study whose entry statistics are not reproduced here. That study reported the block entering at Diebold–Mariano +9.90 against same-engine twins, and reported monthly (+0.75) and per-bar (+1.48) re-estimation of D failing the same gate. Those statistics were computed under a full-sample standardization of the factor scores and are therefore inflated by look-ahead; Section 6.4 supersedes them, and the two design choices they motivated are re-tested there causally. Lag-one D, re-estimated quarterly, is the complete form of the feature. As run on this paper's own battery, the transmission block is the concatenation $[\,G(t) \mid \hat{G}(t)\,]$ — $2q = 40$ columns — under the paper's convention, and $\hat{G}(t)$ alone — $q = 20$ columns — under the documented convention; both conventions are scored at the end of this section.

The width plateau. The width of 20 was inherited, not established by test, so it was varied end to end. The procedure is as follows. The frames are nested: for each width q, the frame is the top-q eigenvectors $v_1, \ldots, v_q$ of the same first-window correlation matrix R, and the full pipeline (scores G, operator D, transmission columns $\hat{G}$, ridge) is re-run at that width. Each width is compared head-to-head against $q = 20$ under the same protocol. The one-bar increment is not flat in q. It improves through $q = 40$ (head-to-head over $q = 20$: +2.29 at daily refit, +4.59 per-bar), holds through $q = 80$, and fails its gate outright at $q = 106$ — every live direction (+1.62 against the twin, the width ladder's first failure). Refinement rungs at $q \in \{30, 50, 60, 70\}$ show the interior is flat to within noise. The forecast improvement therefore sits on a **plateau over widths ~30–80**, with worse performance on both sides: the lead–lag structure occupies a finite-rank sub-span roughly twice the conventional frame, is dense within that sub-span, and is absent beyond it. Three probes locate the mechanism. The widening gain is carried by the new directions' *own* next-bar forecastability, not by new information about the first 20 factors. Doubling the D estimation window rescues nothing at $q = 106$: the tail directions carry no lead–lag content, rather than content estimated too noisily. And the field restricted to the new directions replicates split-half at +0.838 (old block +0.907): the field on the widened span is stable even though the individual eigenvectors inside the span are not identified. The campaign's production configuration runs width 40 as a representative of the plateau, not as a tuned optimum: covering the span is what matters, and the span's truncation point is not identified. All width comparisons here are one-bar results, and the width-40 choice rests on them alone.

6.3 The Four-Block Ridge

The transmission block completes the model. Partition the design at bar t into four column blocks $X_b(t)$, $b \in \{\text{backbone}, \text{exog}, \text{prod}, \text{trans}\}$, with coefficient vectors β_b and per-block penalties α_b. The final specification is one blockwise ridge in a single solver — **719 columns** — refit every bar on a 250-day rolling window $\mathcal{W}$ with a one-day embargo:

$$\hat{\beta} \;=\; \arg\min_{\beta} \sum_{t \in \mathcal{W}} \Big(y_t - \sum_b X_b(t)^\top \beta_b \Big)^2 \;+\; \sum_b \alpha_b \, \|\beta_b\|_2^2,$$

with the four blocks and their penalties

$$y \ \sim\ \text{ridge}_{250\text{d}}\Big[\ \underbrace{\text{target-HAR backbone}}_{27\text{ cols},\ \alpha_{\text{eff}}=1}\ \Big|\ \underbrace{\text{linear exogenous block}}_{516+36\text{ cols},\ 3\times10^{3}}\ \Big|\ \underbrace{\text{frozen products}}_{100\text{ cols},\ 3\times10^{4}}\ \Big|\ \underbrace{\text{transmission}}_{40\text{ cols},\ 3\times10^{3}}\ \Big].$$

[decision pending] The per-block penalties are implemented by column scaling. Scale block b's columns by a constant c_b and solve a single ridge at a common penalty α; writing $\tilde{\beta}_b$ for the coefficients on the scaled columns and substituting $\beta_b = c_b\tilde{\beta}_b$,

$$\sum_{t\in\mathcal{W}}\Big(y_t-\sum_b\big(c_bX_b(t)\big)^{\top}\tilde{\beta}_b\Big)^2+\alpha\sum_b\|\tilde{\beta}_b\|_2^2\ =\ \sum_{t\in\mathcal{W}}\Big(y_t-\sum_b X_b(t)^{\top}\beta_b\Big)^2+\sum_b\frac{\alpha}{c_b^2}\,\|\beta_b\|_2^2,$$

so scaling a block by c_b is identical to penalizing it at $\alpha_b=\alpha/c_b^2$, and one solver call fits all four blocks. The transmission block's own penalty was laddered, and the selected value equals the exogenous block's penalty, not the frozen-product block's.

The increment from the fourth block is small and statistically strong. On the campaign panel, adding the width-40 transmission block to the 699-column per-bar model moves QLIKE from 0.12546 to 0.12529 (Diebold–Mariano +4.59; +2.32 on 2020+), against that panel's HAR baseline of 0.13275. Composed with the second-moment correction of Section 3 in its final form, the campaign's headline number closed at QLIKE 0.12299 ($\approx -7.3\%$ against HAR), having opened the campaign at 0.12579. **[decision pending]** On this paper's own frozen battery, the four-block ridge scores QLIKE 0.22102, with Diebold–Mariano -8.0 against the OLS–HAR incumbent; under the documented convention it scores 0.22650 at -10.5. The transmission *increment* — the paired per-bar comparison of the four-block against the three-block ridge on the same rows — is ΔQLIKE = -0.00009 at DM -1.5 under the stated convention, -0.00002 at -0.4 under the documented one, and -0.00011 at -1.1 under causal per-block tuning, where the tuner is free to place the transmission penalty anywhere in its grid. A fourth variant replaces the frozen-window standardization of the factor scores with trailing standardization over the same window D uses, and passes the resulting columns through the standard rolling scaler — supplying causally the scale stability that the non-causal full-sample standardization of the original study provided illegitimately. Its increment over the three-block ridge is ΔQLIKE = -0.00031 at DM -4.1. These four statistics are the section's verdict on the block.

The trailing-variant increment is not stationary across the sample, and we report its structure rather than only its pooled value. In three eras, the increment per bar is $+12.4\times10^{-5}$ over 2003–2013 ($t=1.78$; the block does not help), -87.7×10^{-5} over 2014–2021 ($t=-5.61$; it helps strongly), and -12.7×10^{-5} over 2022–2024 ($t=-0.77$). The 2014–2021 value is not an outlier artifact: removing the 1% of bars with the largest loss differentials leaves -65.6×10^{-5}, the era median is -13.1×10^{-5}, the per-bar win rate is 52.3%, and every year from 2014 through 2023 has a negative median. Within 2022–2024, 2022 is null (-0.1×10^{-5}), 2023 favors the four-block significantly (-46.6×10^{-5}, $t=-2.29$), and the only positive span is the four-month 2024 stub that ends the panel ($+51.2\times10^{-5}$ over January–April 2024), carried entirely by the factor-score columns rather than the flow columns. The onset of the 2014–2021 era is a step, not a ramp: the rolling twelve-month median of the per-bar differential sits at zero through November 2014 and departs permanently at December 2014–January 2015 (monthly means of -122, -91, -71×10^{-5} for December through February).

No exogenous feed enters the panel in 2013–2015, and the frozen frame assigns any column that was dead in the frame window a loading of exactly zero, so a feed arrival could not enter the block's columns; the one dated change in the panel at that boundary is the homogenization of the equity-aggregate feed cadence, whose last heterogeneous availability bar is December 31, 2014, and which governs aggregate columns that are in the transmission base. That coincidence was tested directly and does not survive. An arm identical to the trailing construction in every respect except that the two cadence-heterogeneous aggregate families are withheld from the transmission base — they remain in the exogenous ridge block, so only what feeds the frame and the operator changes — reproduces the era structure rather than removing it. Within 2003–2013 the two arms are indistinguishable (paired ΔQLIKE $= -3.8 \times 10^{-5}$, DM $t = -0.76$, $p = 0.45$), so the pre-2014 penalty is not attributable to those families; within 2014–2021 the withheld-family arm is significantly *worse* ($+18.9 \times 10^{-5}$, $t = 2.27$, $p = 0.023$), the opposite of what the hypothesis predicts. The December-2014 step is likewise unchanged: measured as the December-2014-through-February-2015 mean minus the September-through-November-2014 mean, it is -72.3×10^{-5} for the incumbent and -74.2×10^{-5} with the families withheld. The per-refresh operator diagnostics agree: across the 76 refreshes the Frobenius norm of the operator is flat across the boundary (pre-2015 mean 0.282 against post-2015 0.279), and refresh-to-refresh drift is *lower* before 2015 than after (0.028 against 0.034), so the operator shows no pre-2015 instability of the kind stale-fill contamination would produce. The cadence-homogenization hypothesis is therefore rejected, and the December-2014 onset is left unexplained. One observation constrains any future explanation: the symmetric-operator variant of Section 6.4 shows essentially no step at that date (-6.2×10^{-5} by the same measure), already sitting at its later level through the autumn of 2014.

The increment is also concentrated in time rather than spread across the 2014–2021 era. Four years account for the entire pooled loss reduction — 2018 contributes 55.6%, 2015 21.1%, 2020 17.3%, and 2017 8.4% — and the remaining years net slightly positive. Per-year Diebold–Mariano statistics reject equal accuracy in only two years (2015, $t = -4.32$; 2018, $t = -4.83$), and the per-bar win rate never exceeds 53% in any era. The concentration is not a small number of outlying bars: trimming the 1% of bars with the largest loss differentials leaves 75% of the era's gain, and within 2015 and 2018 the median differential and the win rate move with the mean. The pooled statistic should be read as evidence that the block helps materially in a few stressed years, not as evidence of a uniform edge.

An ablation decomposes the trailing block's value into its two column groups. With only the trailing-standardized factor scores G (no flow features), the increment over the three-block ridge is -0.00026 at DM -6.0; with only the flow features $\hat{G}$ it is -0.00012 at -2.1; the full block's increment is the -0.00031 at -4.1 reported above. Both components carry signal at the 5% level in the pooled sample; the factor levels carry roughly twice the flow's increment, and the two are partially overlapping. The era structure of the two components differs in kind, not merely in size. The factor-score component helps in every era: -6.9×10^{-5} per bar with $t = -3.36$ already over 2003–2013, -55.9×10^{-5} with $t = -5.70$ over 2014–2021. The flow component changes sign: over 2003–2013 it is significantly harmful ($+21.1 \times 10^{-5}$, $t = +2.95$), and over 2014–2021 it is significantly helpful (-51.0×10^{-5}, $t = -4.60$). The pooled flow increment therefore understates the flow's post-2014 contribution, and the block's era structure is the flow's era structure superimposed on an always-present factor-score contribution.

6.4 Falsification of the Directional Reading

The construction of Section 6.2 retains the antisymmetric part $D = (C - C^{\top})/2$ of the one-bar cross-correlation C precisely because antisymmetry is what encodes direction: $D_{ij} = -D_{ji}$ states that factor i leads factor j exactly as much as j lags i. If the block's forecasting value comes from directional transmission, then replacing D by the symmetric part $S = (C + C^{\top})/2$, which encodes co-movement and carries no direction, should destroy it. Four substitutions test this, each a complete arm on the frozen panel:

1. S in place of D (symmetric control);
2. C itself, undecomposed;
3. the factor scores G alone, with no operator columns at all;
4. the operator columns $\hat{G}$ alone, with no factor scores.

Every comparison below is a paired Diebold–Mariano test on the 248,280 bars common to all arms.

The directional construction is not distinguishable from any of its non-directional controls. Against the three-block ridge, the symmetric control's increment is -0.00026 at DM -2.4, the undecomposed cross-correlation's is -0.00029 at -3.5, and the directional incumbent's is -0.00031 at -4.1. Compared head to head against the directional incumbent on the same rows, the symmetric control loses 4.6×10^{-5} at DM $t = 0.50$ ($p = 0.62$) and the undecomposed operator 1.7×10^{-5} at $t = 0.29$ ($p = 0.78$): neither substitution costs a detectable amount. The factor scores alone — no operator of any kind — give up 5.2×10^{-5} against the full block at $t = 1.00$ ($p = 0.32$), and are statistically indistinguishable from the symmetric control (0.5×10^{-5}, $t = 0.06$, $p = 0.96$). The operator columns alone are the weakest member of the family (-0.00012 at -2.1, losing 18.7×10^{-5} to the full block at $t = 5.25$).

The magnitudes of the three matrices explain the pattern. On the frozen frame, averaged over the 76 quarterly refreshes, $\|C\|_F = 4.73$, $\|S\|_F = 3.20$, and $\|D\|_F = 0.28$: the antisymmetric part carries about 6% of the cross-correlation's energy, and the symmetric part about eleven times the antisymmetric part's. The block's columns are dominated by the symmetric structure whichever operator is nominally applied, and the forecasts behave accordingly.

Two further variants bound the construction's remaining free choices. The frame width K has an interior optimum at the pinned value: the increment improves monotonically from $K = 5$ (-0.00018, DM -4.1) through $K = 10$ (-0.00024, -4.2) to $K = 20$, and collapses at $K = 40$ (-0.00002, -0.2; 21.6×10^{-5} worse than $K = 10$ at $t = 2.62$), so the pinned width is a measured optimum rather than an unexamined constant. Estimating the operator at a two-bar lag instead of one is indistinguishable from the one-bar construction (1.8×10^{-5}, $t = 0.18$). Re-estimating the frame causally from the trailing window at each refresh, rather than freezing it on the first window, removes the effect entirely (-0.00011 at -0.9, not significant): the block's value depends on a fixed coordinate system, not on a current one.

6.5 The Best Pooled Model

The falsification result has a direct consequence for the model the paper carries forward. If the operator columns contribute nothing detectable, the block should be built without them. The arm that does so — per-block causal tuning, trailing standardization, and a transmission block containing only the 20 factor-score columns — scores QLIKE 0.21909 at DM -13.7 against the benchmark, with an increment of -0.00033 at -3.1 over the tuned three-block ridge. It is the best pooled model among the constructions this section presents at $K = 20$. A wider frame does better still, and a control settles what the gain belongs to. Doubling the frame to $K = 40$ while leaving the penalty flat scores 0.21894 (DM -13.4), improving on the $K = 20$ arm by -0.00015 at -1.7 and on the tuned three-block by -0.00049 at -4.1. Running the same $K = 40$ frame with a rank-shaped rather than flat penalty changes nothing measurable (+0.00003 at +0.5). The frame width is the lever; the penalty shape is not. The headline model is therefore the wider frame under a flat penalty — the simpler of the two constructions, with no shape parameter to select. Against the corresponding full-block model it is statistically indistinguishable (-0.00006 at -0.5), so the two are equivalent on accuracy and the levels-only construction is preferred on parsimony: half the columns, and none of the operator machinery — no trailing estimation window, no refresh schedule, no lag choice. The full-block figures are reported next for comparison, and the master table of Appendix A carries both.

Combining the trailing construction with per-block causal tuning yields the full-block counterpart: the tuned trailing-transmission four-block ridge scores 0.21915 (DM -13.3 against the benchmark), with an increment of -0.00028 at -2.1 over the tuned three-block.

The transmission block is the only block in the model that uses information at a delay: the other three read the current bar, and this one predicts next-bar factor motion from the previous bar. On a cross-section of assets, the same construction — a lead–lag field estimated across names rather than across one market's factors — becomes the estimation target rather than an added feature block. That extension lies beyond this paper.

7 Conclusion

[decision pending]

A Master Table of Campaign Arms

Table 5 lists every arm of the unification campaign: one frozen panel (Section 2), one scoring contract (Section 3), every arm scored on the identical rows. QLIKE is the pooled contract loss; Δ is against the benchmark; DM is the Diebold–Mariano t-statistic of the per-bar loss differential against the benchmark. Product- and transmission-block arms with a 24,000-bar window are scored from the first row outside their frozen selection and frame windows.

Table 5: All campaign arms under the frozen panel and scoring contract.

Model	QLIKE	Δ	DM t
Benchmark			
OLS on HAR + calendar (benchmark)	0.23353	—	—
Single-bucket attribution (minimum-norm least squares)			
HAR + moments bucket	0.23089	-0.00264	-5.7
HAR + liquidity bucket	0.23471	+0.00118	+3.3
HAR + market bucket (equal-weighted)	0.23357	+0.00004	+0.1
HAR + market bucket (value-weighted)	0.23303	-0.00050	-1.2
HAR + sentiment bucket	0.23253	-0.00100	-5.2
HAR + implied-volatility bucket	0.23363	+0.00010	+0.4
HAR + volatility-demand bucket	0.23375	+0.00022	+1.2
HAR + all exogenous columns (joint)	0.23444	+0.00091	+1.6
Bucket designs under causally tuned ridge			
HAR + moments bucket	0.23089	-0.00264	-3.6
HAR + liquidity bucket	0.23431	+0.00078	+0.8
HAR + market bucket (equal-weighted)	0.23354	+0.00001	+0.0
HAR + market bucket (value-weighted)	0.23296	-0.00057	-0.9
HAR + sentiment bucket	0.23357	+0.00004	+0.1
HAR + implied-volatility bucket	0.23362	+0.00009	+0.2
HAR + volatility-demand bucket	0.23397	+0.00044	+0.7
HAR + all exogenous columns (joint)	0.23040	-0.00313	-4.4
Bucket designs under causally tuned elastic net (free mixing)			
HAR + moments bucket	0.23134	-0.00219	-2.5
HAR + liquidity bucket	0.23453	+0.00100	+0.9
HAR + market bucket (equal-weighted)	0.23382	+0.00029	+0.4
HAR + market bucket (value-weighted)	0.23361	+0.00008	+0.1
HAR + sentiment bucket	0.23389	+0.00036	+0.5
HAR + implied-volatility bucket	0.23401	+0.00048	+0.6
HAR + volatility-demand bucket	0.23422	+0.00069	+0.9
HAR + all exogenous columns (joint)	0.23116	-0.00237	-2.0
Penalized linear, full design			
Ridge, fixed $\alpha = 0.1$	0.23361	+0.00008	+0.1
Ridge, fixed $\alpha = 0.3$	0.23312	-0.00041	-0.8
Ridge, fixed $\alpha = 1$	0.23243	-0.00110	-2.1
Ridge, fixed $\alpha = 3$	0.23169	-0.00183	-3.6
Ridge, fixed $\alpha = 10$	0.23087	-0.00266	-5.4
Lasso, fixed $\alpha = 10^{-4}$	0.22950	-0.00402	-8.4
Ridge, causally tuned	0.23040	-0.00313	-4.4
Elastic net, causally tuned	0.23056	-0.00297	-2.6
Lasso, causally tuned	0.23134	-0.00219	-2.0
Block ridges			
Two-block ridge (stated penalties)	0.22941	-0.00412	-9.0
Three-block ridge (stated penalties)	0.22111	-0.01242	-7.7
Four-block ridge (stated penalties)	0.22102	-0.01251	-8.0
Two-block ridge (documented penalties)	0.22775	-0.00578	-9.6
Three-block ridge (documented penalties)	0.22652	-0.00701	-10.6
Four-block ridge (documented penalties)	0.22650	-0.00703	-10.5
Two-block ridge (per-block causal tuning)	0.22822	-0.00531	-11.7
Three-block ridge (per-block causal tuning)	0.21942	-0.01411	-12.8

B Smear-Convention Sensitivity, All Arms

Table 6 reports every completed arm's pooled QLIKE under the three back-transform conventions of Section 3, with within-convention ranks. The rank agreement statistics quoted there (τ against the contract ranking: none 0.637, traditional Duan 0.918) are computed over this table.

Table 6: Pooled QLIKE per arm under the three back-transform conventions (within-convention rank in parentheses).

Arm
OLS on HAR + calendar (benchmark)
HAR + moments bucket (OLS)
HAR + liquidity bucket (OLS)
HAR + market EW bucket (OLS)
HAR + market VW bucket (OLS)
HAR + sentiment bucket (OLS)
HAR + implied vol bucket (OLS)
HAR + vol demand bucket (OLS)
HAR + all features bucket (OLS)
Ridge, fixed $\alpha = 1$
Lasso, fixed $\alpha = 10^{-4}$
Ridge, causally tuned
Lasso, causally tuned
Elastic net, causally tuned
HAR + moments bucket (ridge, causally tuned)
HAR + liquidity bucket (ridge, causally tuned)
HAR + market EW bucket (ridge, causally tuned)
HAR + market VW bucket (ridge, causally tuned)
HAR + sentiment bucket (ridge, causally tuned)
HAR + implied vol bucket (ridge, causally tuned)
HAR + vol demand bucket (ridge, causally tuned)
HAR + all features bucket (ridge, causally tuned)
HAR + moments bucket (elastic net, causally tuned, free mixing)
HAR + liquidity bucket (elastic net, causally tuned, free mixing)
HAR + market EW bucket (elastic net, causally tuned, free mixing)
HAR + market VW bucket (elastic net, causally tuned, free mixing)
HAR + sentiment bucket (elastic net, causally tuned, free mixing)
HAR + implied vol bucket (elastic net, causally tuned, free mixing)
HAR + vol demand bucket (elastic net, causally tuned, free mixing)
HAR + all features bucket (elastic net, causally tuned, free mixing)
HAR + moments bucket (elastic net, causally tuned, free mixing, extended penalty grid)
HAR + all-features bucket (elastic net, causally tuned, free mixing, extended penalty grid)
Ridge, fixed $\alpha = 0.1$
Ridge, fixed $\alpha = 0.3$
Ridge, fixed $\alpha = 3$
Ridge, fixed $\alpha = 10$
Two-block ridge (stated penalties)
Three-block ridge (stated penalties)
Four-block ridge (stated penalties)
Two-block ridge (documented penalties)
Three-block ridge (documented penalties)
Four-block ridge (documented penalties)
Two-block ridge (per-block causal tuning)
Three-block ridge (per-block causal tuning)
Four-block ridge (per-block causal tuning)
Four-block ridge (trailing-standardized transmission)
Four-block ridge (trailing, factor levels only)
Four-block ridge (trailing, lead-lag flow only)
Four-block ridge (trailing transmission, per-block causal tuning)

References

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Naihua Duan. Smearing estimate: A nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383):605–610, 1983.

Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.

Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Yin Lou, Rich Caruana, Johannes Gehrke, and Giles Hooker. Accurate intelligible models with pairwise interactions. In *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 623–631, 2013.

Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996.